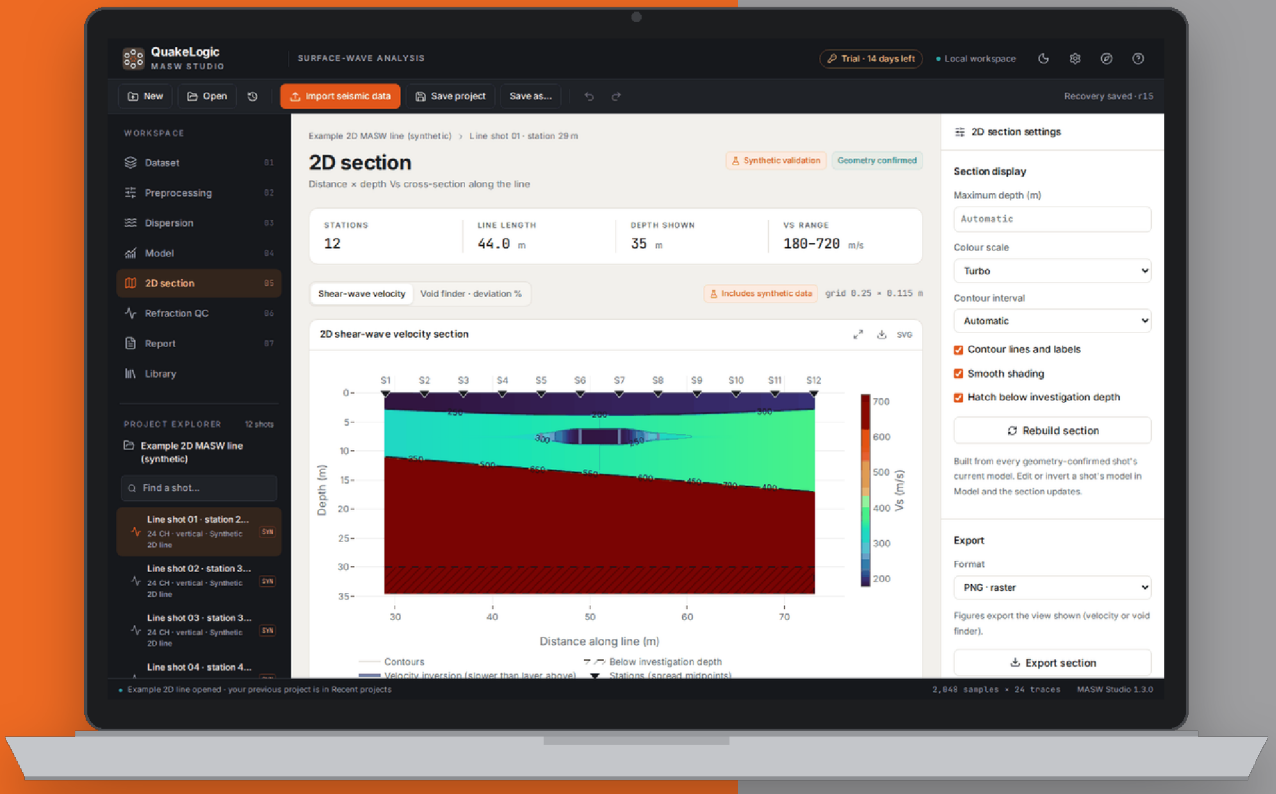
QUAKELOGIC

# QUAKELOGIC

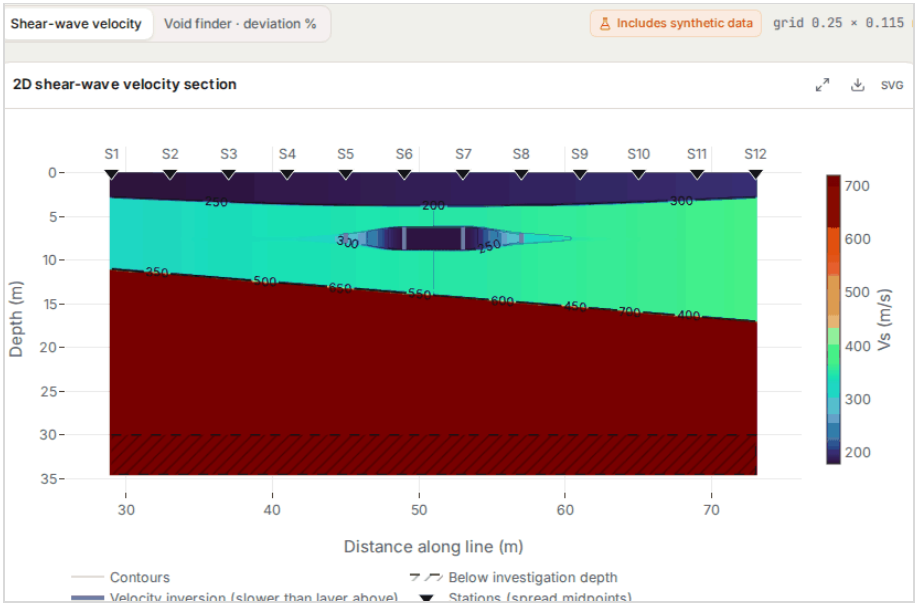
## SURFACE-WAVE ANALYSIS SOFTWARE

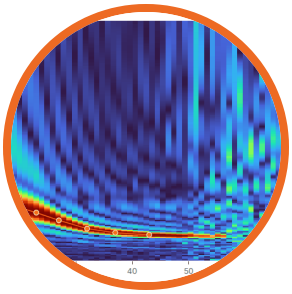
MASW STUDIO

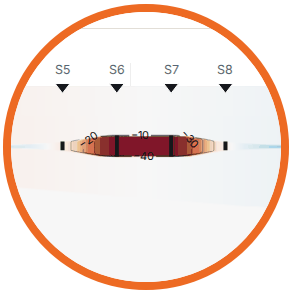


# MASW STUDIO

## MULTICHANNEL ANALYSIS OF SURFACE WAVES · WINDOWS DESKTOP






## FROM FIELD RECORDS TO VS30 AND 2D SECTIONS

QuakeLogic MASW Studio is a standalone Windows application for active-source MASW surveys. It takes multichannel shot records from the seismograph through preprocessing, dispersion imaging and picking, forward modelling and bounded inversion to a layered shear-wave velocity profile, Vs30, a 2D Vs section of the whole line and a technical PDF report. It runs entirely offline, records every processing step, and never modifies the raw field data.

- Reads field data directly: SEG-2, SEG-Y, DRM (.drm) and text exports, recognised from the file contents, a whole survey line at once.
- Defensible results: every step and parameter is recorded in the project, and Vs30 is reported only within the declared interpretation depth.
- The 2D void finder shows each point against its layer average along the line, flagging slower ground, loose fill and possible voids in red.
- Fully offline with no account or telemetry: projects, recovery copies and licences stay on your own computer, ready for remote field sites.

## STANDARD FEATURES »

- SEG-2, SEG-Y, DAT, DRM import
- Geometry review and confirmation
- Butterworth filters
- Time and linear mutes
- Phase-shift F–V imaging
- F–K spectrum
- Aliasing and aperture QC
- Click-to-pick named modes
- Rayleigh and Love modelling
- Bounded least-squares inversion
- Vs30 average line (0–30 m)
- 2D Vs section of the line
- Void finder deviation view
- Refraction first-arrival QC
- PDF report with live preview
- PNG, SVG, PDF and CSV export
- Library of past analyses
- Guided tour and example data



QUAKELOGIC

# KEY FEATURES

- **Direct field-data import.** SEG-2 (including 20-bit), SEG-Y, DRM (.drm) and time-column text, with geometry and units read from the headers and confirmed before analysis.
- **Transparent preprocessing.** Detrend, zero-phase Butterworth filters and mutes are previewed beside the original record, then applied as a recorded, removable sequence.
- **Dispersion imaging.** Phase-shift F–V and F–K transforms with phase-coherence normalisation, plus aperture and spatial-aliasing guides showing what the array can resolve.
- **Mode picking.** Click to trace the fundamental and higher modes as named curves, each with explicit uncertainty carried into the inversion.
- **Modelling and inversion.** Rayleigh and Love modes of a layered elastic earth and a bounded, uncertainty-weighted inversion with full convergence history.
- **2D section and void finder.** Station models along a roll-along line become a smooth distance–depth Vs section, with investigation depth hatched and velocity inversions flagged.
- **Reporting.** A branded technical PDF with survey details, site photos, figures and tables, previewed page by page before saving.
- **Library.** Every analysis on the computer is searchable by client, site or notes, and can be reopened or merged into a new project.

## PRODUCT FEATURES

**Data import**
SEG-2 rev 1 · SEG-Y rev 0–2 · DRM (.drm) · CSV/TXT/DAT text · up to 512 channels

**Processing**
Detrend · Butterworth low/high/band-pass · time and linear mutes · trace edit, stack, split

**Dispersion**
Phase-shift F–V · F–K · coherence or energy normalisation · acquisition QC guides

**Modelling**
Rayleigh (P–SV) and Love (SH) modes · multi-mode forward curves · model comparison

**Inversion**
Bounded trust-region least squares · fixed Vp and density · convergence history

**Interpretation**
Vs30 travel-time average · declared interpretation depth · 2D Vs and deviation sections

**Output**
Technical PDF report · PNG/SVG/PDF figures · CSV tables · self-contained .qlmasw projects

**For more information about the features, specifications, and available configurations of the MASW STUDIO, please contact QuakeLogic.**

## SYSTEM REQUIREMENTS & DELIVERY

| COMPONENT | CONFIGURATION |
|---|---|
| Operating system | Windows 10 (1809 or later) or Windows 11, 64-bit |
| Memory | 8 GB RAM recommended |
| Disk space | About 1 GB |
| Display | Resizable window; light and dark themes |
| Internet connection | Not required; runs fully offline |
| Installer | One installer with bundled runtime |
| Projects | Self-contained .qlmasw files with raw data and history |
| Recovery | Automatic local recovery copies, undo and redo |
| Licence | Per computer, offline signed licence file |
| Trial | 14 days, full function, outputs marked TRIAL |
| Updates | One year of updates included, renewable |
| Support | support@quakelogic.net · reply within 3 business days |

## SUPPORTED FIELD DATA

| SPECIFICATION | VALUE |
|---|---|
| SEG-2 | .sg2 · .seg2 · .seg · .dat; 16/32-bit integer, 20-bit, IEEE float |
| SEG-Y | .sgy · .segy · .dat; IBM/IEEE float, 8/16/32-bit integer |
| DRM recordings | .drm field recordings with header geometry |
| Text columns | .csv · .txt · .dat · .asc; time in s or ms |
| Channels per shot | 1 to 512 |
| File size | Up to 256 MiB per record |
| Samples | Up to 32 million per shot |
| Geometry units | Metres or feet, converted and confirmed |
| Survey lines | Multi-shot import for roll-along 2D lines |

# OPTIONAL FEATURES & ACCESSORIES

### PERPETUAL LICENCE

Use the licensed versions on one computer indefinitely, identified by its Machine ID and activated offline with a signed licence file.

### MULTI-SEAT & ACADEMIC

Licences for several computers, teaching laboratories and research groups, issued as one licence file covering every Machine ID.

### MAINTENANCE RENEWAL

Extends access to new versions, new import formats and analysis features for another year after the included first year.

### FIELD ACQUISITION PACKAGE

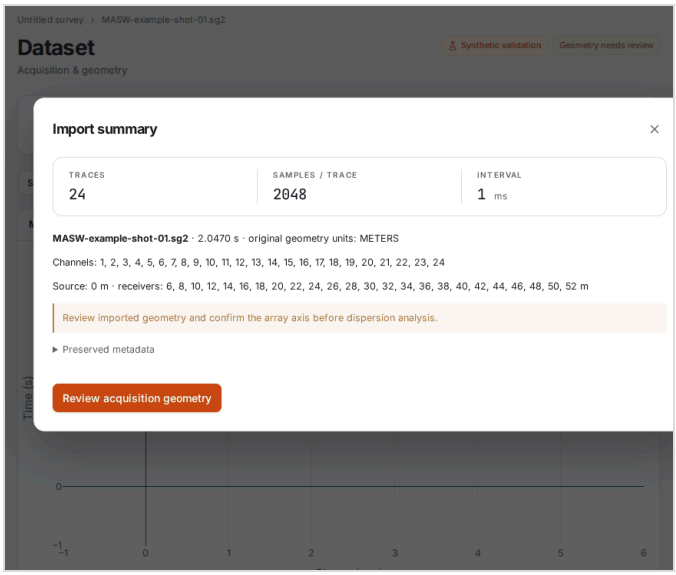
Pair with a QuakeLogic seismograph and geophones for a complete MASW acquisition and analysis package from one supplier.

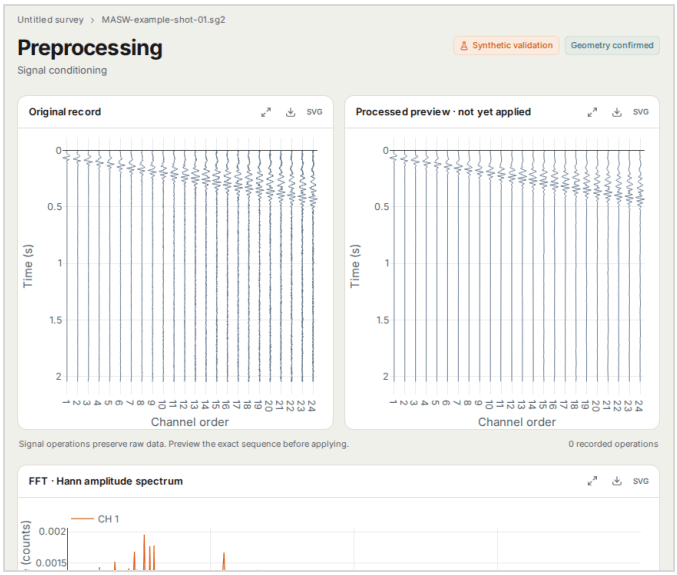
## TECHNICAL SPECIFICATIONS

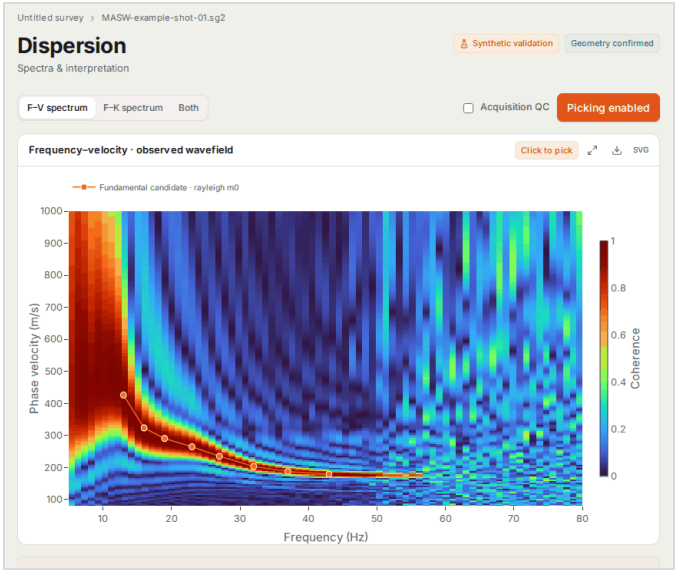
| SPECIFICATION | DESCRIPTION |
|---|---|
| Application | Native Windows desktop window |
| Architecture | 64-bit, bundled runtime |
| Network use | None; no ports, no cloud |
| Telemetry | None |
| Max channels | 512 per shot |
| Max record | 256 MiB, 32 million samples |
| Import formats | SEG-2, SEG-Y, DRM, CSV/TXT/DAT |
| SEG-2 encodings | Codes 1–5 incl. 20-bit |
| Filters | Butterworth order 1–10, zero-phase or causal |
| Mutes | Time window and linear |
| F–V transform | Phase shift (Park et al. 1999) |
| Normalisation | Phase coherence or energy |
| F–K transform | 2D spatial Fourier |
| QC guides | Aperture and spatial aliasing |
| Picking | Named modes with uncertainty |
| Wave types | Rayleigh (P–SV), Love (SH) |

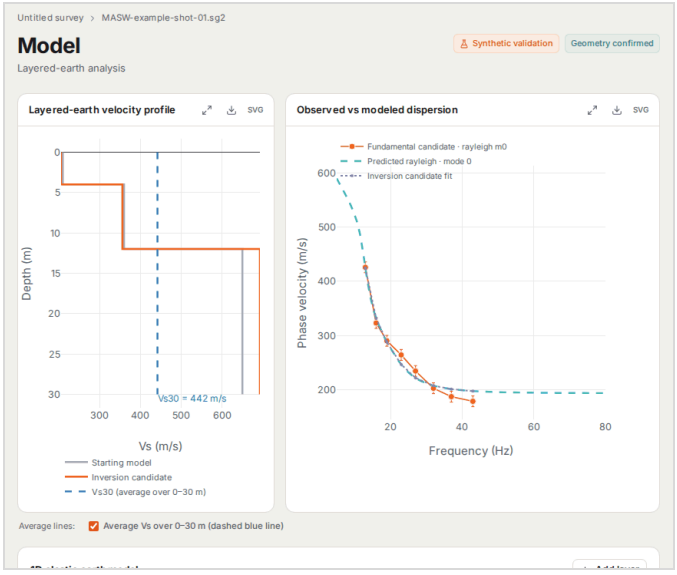
| SPECIFICATION | DESCRIPTION |
|---|---|
| Forward solver | Dunkin / Thomson–Haskell |
| Inversion | Bounded trust-region least squares |
| Model | Layered Vs; Vp and density fixed |
| Vs30 | Travel-time average, 0–30 m |
| 2D section | Shape-preserving interpolation |
| Void finder | Deviation from layer mean, % |
| Section QC | Investigation depth hatching |
| Refraction QC | First-arrival picks and velocities |
| Report | Branded PDF with page preview |
| Figure export | PNG, SVG and PDF |
| Table export | CSV |
| Project file | .qlmasw, checksummed |
| History | Undo, redo, recovery copies |
| Guidance | Guided tour, example SEG-2, user guide |
| Licensing | Offline signed licence per computer |
| Trial | 14 days, outputs marked TRIAL |

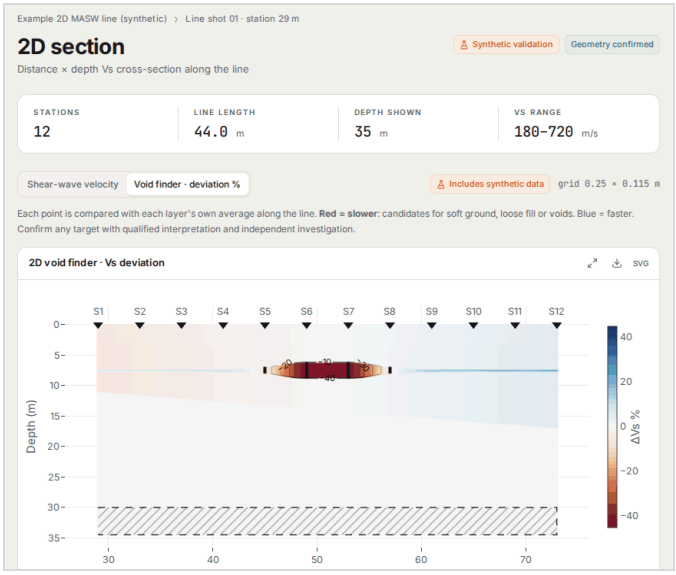
# PRODUCT GALLERY

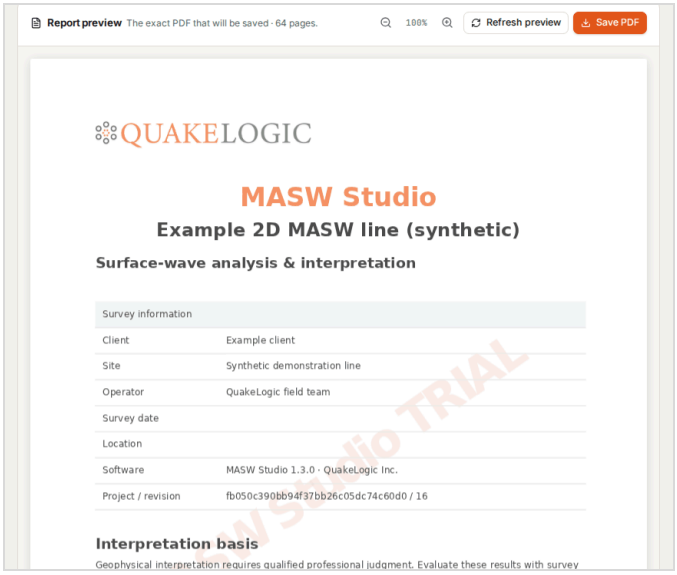


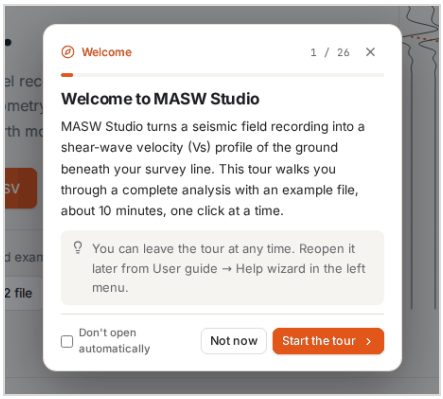



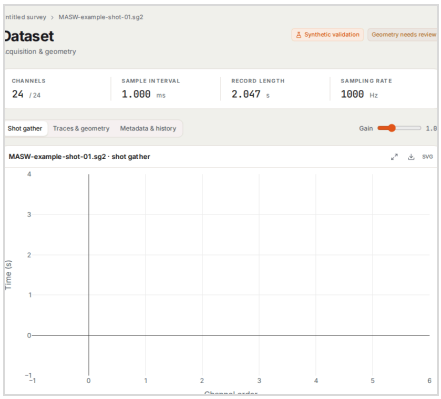



QUAKELOGIC

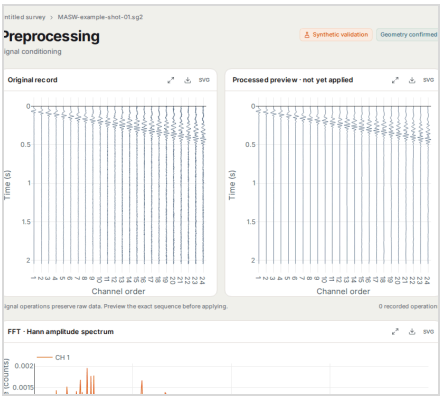
# APPLICATION WORKSPACES



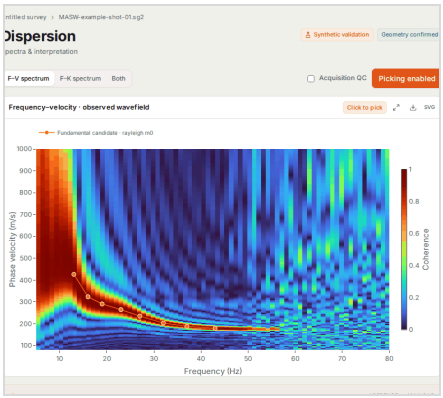

GUIDED TOUR



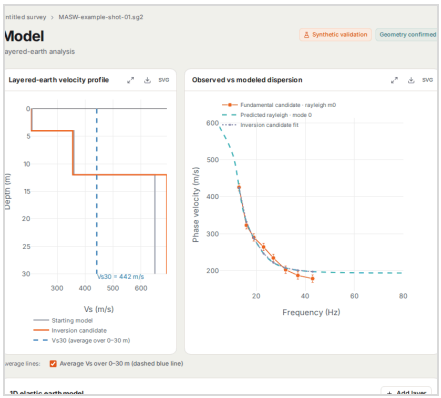

SHOT GATHER



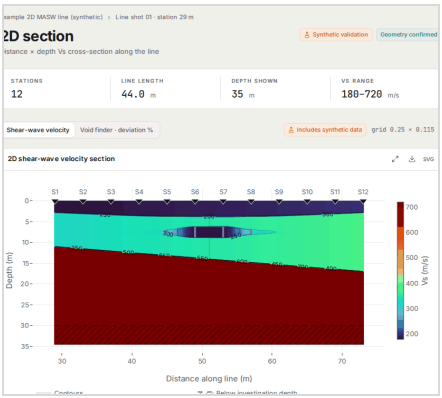

FILTER PREVIEW



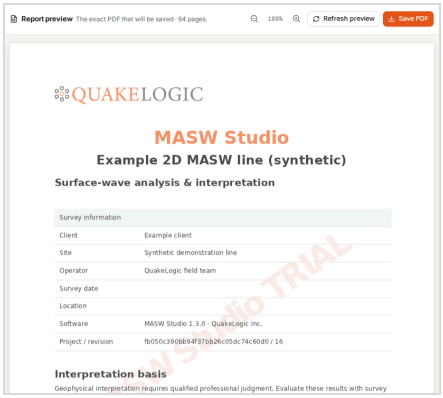

MODE PICKING



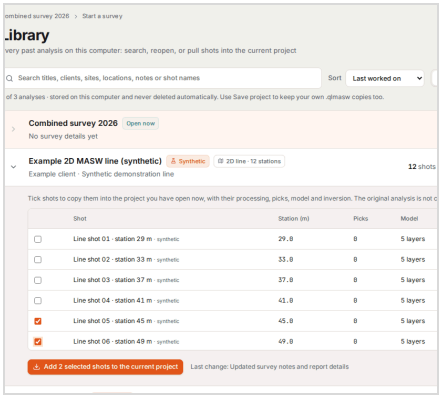

VS PROFILE & VS30



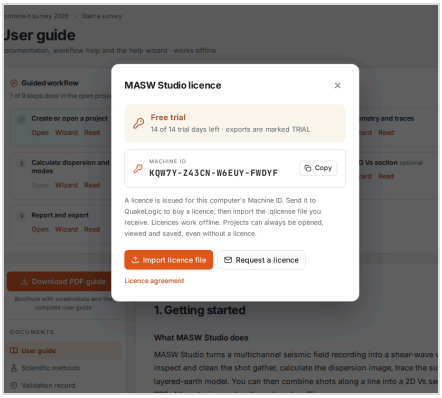

2D VS SECTION




PDF REPORT




ANALYSIS LIBRARY




OFFLINE LICENSING

QUAKELOGIC

MASW STUDIO

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**Email:** sales@quakelogic.net
**Web:** products.quakelogic.net

## About QuakeLogic:

QuakeLogic Inc. is a California-based engineering and technology company providing innovative, reliable, high-quality solutions to customers worldwide.

## Our Commitment

We are committed to quality, innovation, and customer satisfaction — building lasting partnerships through reliable solutions and dependable support.

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